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COMPUTER SECURITY WK3

1.SCENARIO ANALYSIS:

Risk Management Scenario: Web Application Security

A company operates an online shopping website that stores customer names, addresses, passwords, and payment information. The company faces risks such as hacking, phishing, malware, data breaches, and unauthorized access. To manage these risks, the company develops a web application security policy that requires strong passwords, multi-factor authentication, regular software updates, secure data encryption, and controlled access to sensitive information. Employees are trained to recognize phishing emails and other cyber threats. The company also performs regular vulnerability assessments and security testing to identify weaknesses before attackers can exploit them. Firewalls and intrusion detection systems are used to monitor suspicious network activity. In the event of a security breach, an incident response plan is followed to contain the attack, recover affected systems, and notify customers when necessary. These measures reduce the likelihood of successful attacks, protect customer information, maintain business operations, and help the company comply with data protection and cybersecurity requirements.

2.CONCEPT RESEARCH:

NIST Risk Management Framework

The NIST Risk Management Framework (RMF) is a structured approach developed by the National Institute of Standards and Technology to help organizations manage cybersecurity and information security risks. Its role is to identify, assess, and control risks affecting systems, data, and users. The framework guides organizations through important activities such as preparing systems, categorizing information, selecting security controls, implementing controls, assessing their effectiveness, authorizing systems, and continuously monitoring them. By following the NIST RMF, organizations can make informed security decisions, reduce vulnerabilities, protect sensitive information, and respond effectively to changing cyber threats. It also supports accountability and helps organizations maintain secure and reliable information systems.

3.TOOL PRACTICE:

[Download the Risk Matrix Excel file](#)

Creating a risk matrix in Excel helped me understand how organizations identify and manage potential security risks. I listed different risks, considered their likelihood and impact, and used these factors to determine the overall risk level. This made it easier to identify which risks require immediate attention and which ones can be monitored. The activity also showed me the importance of having clear security controls and preventive measures. Using Excel made the information organized and easy to update. Overall, the exercise improved my understanding of risk assessment and demonstrated how a risk matrix can support better decision-making and strengthen an organization's security.

4.DIAGRAM:

The risk management workflow diagram shows the main steps an organization follows to manage security risks. First, risks are identified by finding possible threats and vulnerabilities. Next, the risks are assessed by considering their likelihood and potential impact. The organization then evaluates and prioritizes the risks based on their severity. After that, appropriate controls and security measures are implemented to treat or reduce the risks. The organization documents the actions taken and communicates important information to relevant stakeholders. Finally, risks are regularly reviewed and the process is improved. Continuous monitoring helps identify new risks and ensures that security controls remain effective.

INFORMATION SECURITY POLICY WORKFLOW

A Continuous Process for Managing Risks and Ensuring Compliance

